

Genetic report for longevity

# Maximum Longevity - Genetic Synthesis

Centenary variants and life expectancy still to be read (losses of function of APOC3, ANGPTL3/4/8 and PCSK9, GHR d3, SIRT6 of centenarians, TOMM40, CDKN2A/B), longevity scores and synthesis of genetic reports: map of 84 mechanisms, points of greatest impact and what the chip does not see

Sérgio F · Report date: 03-10-2026

Last report of the series. Part A reads the variants of centenarians and longevity that still lacked: FOXO3 (4 SNPs), CETP I405V, the loss of protective function of APOC3, ANGPTL3/4/8 and PCSK9 (with screening of all ClinVar in these genes), KL-VS of klotho, APOE with TOMM40/APOC1, IGF1R, GHR d3, SIRT6 (informative), 9p21 (CDKN2A/B), life expectancy locus of GWAS and the scores of longevity and parental life expectancy. Part B synthesizes all genetic reports integrated in the profile: the genetic status of each 84 aging mechanisms of the target list, the 15 unfavourable points with the greatest impact (each with the evidence and the main suggestion), the 10 more favourable points and what a chip cannot see (complete genome and clonal hematopoiesis).

Only genes: genotyping chip + local imputation + polygenic scores (PGS Catalog) and loci scores of GWAS Catalog, with percentiles in the reference population. Each theme is divided into favourable and unfavourable and linked to the aging mechanisms of the list of objectives. It is neither a diagnosis nor a prescription. All practical ideas are marked as a suggestion (not applied).

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## Summary

**42**

variants

**30**

in the chip

**8**

imputed  
( $DR2 \geq 0.8$ )

**4**

not found

**5**

Scores with  
percentile

**Overall reading.** Part A. FOXO3 is homozygous for the 4 alleles of longevity (rs2802292 G/G, rs2764264 C/C, rs13217795 C/C, rs2802288 A/A), the most favourable finding of this part. APOE  $\epsilon 3/\epsilon 3$  with typical TOMM40; without KL-VS (typical klotho); GHR with 1 copy d3 and IGF1R typical; SIRT6 rs107251 is only informative and the centenary variant (centSIRT6) is not read. None of the protective function losses of lipids is present (APOC3 R19X and other 3, ANGPTL4 E40K, PCSK9 R46L, Y142X and C679X), as in more than 95% of people; from the screening of 343 variants of ClinVar in these genes, 19 were read and none is present. CETP is of high activity (3 SNPs, a single signal; weak unfavourable). 9p21 is homozygous for the risk haplotype in the 5 SNPs of the haplotype (4 in the chip: rs1333049 C/C, rs10757278 G/G, rs4977574 G/G and rs1333042 G/G; 1 imputed: rs1556516 C/C,  $DR2 = 1$ ), a single signal confirmed by linkage imbalance in the reference population ( $r = 0.92-0.998$ ): the strongest unfavourable point. In the scores, longevity (330 variants) is in the 92nd percentile and the score of parental life expectancy loci in the 16 (less alleles that shorten it): the two favourable. Part B. Of the 84 mechanisms, 15 have favourable genetic status, unfavourable 10, mixed 21 and typical 9; 28 does not have possible genetic reading with chip (mainly core, RNA, fine proteostasis and peroxisomes) and 1 is out of scope by rule. The profile is of good polygenic foundations with focal weaknesses. The 3 points with the greatest impact are homozygous 9p21, liver (PiMZ + PNPLA3) and mental health (depression, ADHD and insomnia); venous thrombosis and pulmonary embolism, aorta and aortic valve, low lean mass, small cerebral vessels and colorectal cancer (complete scores, in the 94.3 and 78.5 percentiles, correct the typical reading of small scores). The most recent reports add the tendency for bacterial mucosal infections (pneumonia in 84, urinary infection in 94) and for tinnitus (loci in 99.8), with typical presbycusis and severe COVID-19 in 4.7. The most favourable 3 are APOE  $\epsilon 3/\epsilon 3$  with MAPT H2/H2, FOXO3 with low IGF-1 axis and very low polygenic coronary risk. The chip does not see rare variants (only 21% of the pathogenic variants known in the ACMG genes are read), GSTM1/GSTT1, complete HLA, repetitions such as Lp(a), or acquired clonal hematopoiesis: the two measurements with more value to add are a complete genome to 30x and a panel of CHIP in the blood.

### Favourable

- **Longevity (330 variants):** the 92nd percentile
- **Parental life expectancy (GWAS loci):** the 16th percentile
- **FOXO3 rs2802292 GG:** FOXO3 rs2802292: allele G = longevity (Willcox 2008; meta-analyzes in several populations).
- **FOXO3 rs2764264 CC:** FOXO3 rs2764264: C allele = longevity (Willcox 2008; Flachsart 2009).
- **FOXO3 rs13217795 CC:** FOXO3 rs13217795: C allele : longevity.
- **FOXO3 rs2802288 AA:** FOXO3 rs2802288: allele A = longevity (Flachsart 2009).

### Unfavourable

- **CETP (I405V) rs5882 AA:** CETP I405V: allele V (G) low to CETP and was more frequent in a cohort of centenarians (Barzillai 2003); inconsistent replication.
- **CDKN2B-AS1 (9p21) rs1333049 CC:** 9p21 rs1333049: the allele C is the coronary risk (the strongest coronary sign of GWAS).
- **CDKN2B-AS1 (9p21) rs10757278 GG:** 9p21 rs10757278: the allele G is that of coronary risk; same block.
- **CDKN2B-AS1 (9p21) rs1333042 GG:** 9p21 rs1333042: allele A was associated with more longevity of the father (Wright 2019); same block.
- **CDKN2B-AS1 (9p21) rs1556516 CC:** 9p21 rs1556516: parental life expectancy locus (Pilling 2017); the C allele follows the coronary risk of rs1333049 C (same haplotype).

Here, the scores out of the typical range ( $\leq 25$  or  $\geq 75$ , in the trait direction) and the variants with strong or moderate evidence. The weak evidence variants are in the tables of each theme.

**Strand:** All genotypes are in the strand + (forward) of GRCh38, the same as dbSNP/Ensembl. Of the 30 chip genotypes, 0 needed complement, 0 are palindromic (A/T or C/G; orientation confirmed by the position and reference alleles of the chip, which is already in the strand +) and 0 were read by the position (merged rsID). The imputations come from the local VCF, also in the strand +. The cited effect alleles of studies that use the strand of the gene were converted to the strand +; therefore, some genotypes appear with different letters from those in the literature (e.g., genes in the strand – such as COL1A1 or VDR).

**Polygenic scores update (03-10-2026).** The scores were recalculated with more published recent scores (the same method and the same reference population). This box prevails over the previous values cited in the text.

- Coronary disease: new scores p54, **p18** and p5, 5; with the previous ones, the consensus is p18. It remains low, less extreme than p5.
- Lean mass: new p32 (genomic) and p3,5 scores; with the previous ones (**p27**, p30 and p10), the consensus is p27. Under the average, lower than the p10 cited; strength and protein training remain.
- Fractures: the only new score (45 SNPs) gives p16, against p84 in the previous one; the genetic risk of fracture becomes uncertain and bone densitometry decides.
- Alzheimer's disease: the new genomics give p75 and p85 (before p52); with APOE ε3/ε3, reading goes from medium to medium-high (genomic consensus p75). The suggestions remain.
- Confirmed: high colorectal (p80), prostate (p5,5), melanoma (p18) and low atrial fibrillation (p9), depression (p85), glaucoma (p85) and high insomnia (p80), typical 2 diabetes (p58).

## FOXO3, GH/IGF-1 axis and sirtuins

FOXO3 is the second most replicated gene of human longevity, after APOE: the haplotype that includes the G allele of rs2802292 was described in a cohort of men older than 95 years (Willcox 2008) and replicated in several cohorts. Insulin, IGF-1 and growth hormone (GH) signaling is the most conserved longevity axis among species: less signal (IGF1R with less function, GH receptor without exon 3) is associated with longer life in animals and, weakerly, in humans. GHR d3 (exion 3) is read by rs6873545 (allele C = d3; almost perfect marker of deletion); in men from various cohorts, d3/d3 was associated with about 10 years of life (Ben-Avraham 2017). The IGF1R A37T variant (rs121912426) was found in centenarians with partial resistance to IGF-1 (Suh 2008). CentSIRT6 (N308K + A313S, rs201141490 and rs183444295) is a rare variant of SIRT6 enriched in centenaries that improves the repair of DNA and stops backtransposions (Simon 2022); only informative. The FOXO3, the IGF1R and the rs107251 of the SIRT6 have already been read in the Exceptional Longevity report (profile §45); here are confirmed and joined.

**Ageing mechanisms:** Deregulation of Nutritional Sensing; Insulin resistance; Endocrine dysregulation; Proteostasis loss; Autophagy / Deregulated Macroautophagy

### Favourable

- FOXO3 rs2802292 GG
- FOXO3 rs2764264 CC
- FOXO3 rs13217795 CC
- FOXO3 rs2802288 AA

### Unfavourable

None.

### Favourable variants (4)

| Gene         | Variant                           | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                              |
|--------------|-----------------------------------|-----------|--------|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------|
| <b>FOXO3</b> | rs2802292<br>6:108587315 · G/T    | <b>GG</b> | chip   | T (0; 57%)                                            | moderate | FOXO3 rs2802292: allele G = longevity (Willcox 2008; meta-analyzes in several populations). |
| <b>FOXO3</b> | rs2764264<br>6:108613258 · C/G/T  | <b>CC</b> | chip   | T (0; 63%)                                            | moderate | FOXO3 rs2764264: C allele = longevity (Willcox 2008; Flachsbart 2009).                      |
| <b>FOXO3</b> | rs13217795<br>6:108652895 · C/G/T | <b>CC</b> | chip   | T (0; 63%)                                            | moderate | FOXO3 rs13217795: C allele : longevity.                                                     |
| <b>FOXO3</b> | rs2802288<br>6:108575012 · A/G/T  | <b>AA</b> | chip   | G (0; 57%)                                            | moderate | FOXO3 rs2802288: allele A = longevity (Flachsbart 2009).                                    |

### Typical / intermediate variants (3)

| Gene                          | Variant                          | Genotype  | Source                 | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                                                              |
|-------------------------------|----------------------------------|-----------|------------------------|-------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GHR (d3)</b>               | rs6873545<br>5:42631162 · T/C    | <b>CT</b> | chip                   | T (1; 70%)                                            | weak     | GHR d3: the C allele marks the exon deletion 3; d3/d3 was associated with about 10 years of life in men (Ben-Avraham 2017); 1 copy has intermediate effect. |
| <b>GHR (d3, confirmation)</b> | rs4590183<br>5:42626104 · T/A/C  | <b>CT</b> | imputation<br>DR2=1.00 | T (1; 70%)                                            | weak     | Second marker of exon deletion 3 (allele C); confirms the reading of rs6873545.                                                                             |
| <b>IGF1R</b>                  | rs2229765<br>15:98934996 · G/A/T | <b>AG</b> | chip                   | G (1; 57%)                                            | weak     | IGF1R rs2229765: Allele A is associated with lower IGF-1 and was more frequent in long-lived (Bonafè 2003).                                                 |

## Non-judgmental variants (informative) (2)

| Gene                | Variant                            | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                              |
|---------------------|------------------------------------|-----------|--------|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------|
| <b>IGF1R (A37T)</b> | rs121912426<br>15:98707880 · G/A/T | <b>GG</b> | chip   | —                                                     | weak     | IGF1R A37T: rare variant described in centenarians (Suh 2008); absent (as almost everyone). |
| <b>SIRT6</b>        | rs107251<br>19:4176088 · T/C       | <b>CT</b> | chip   | —                                                     | weak     | SIRT6 rs107251: weak association with survival in the elderly; only informative.            |

## SNP not found in the data (2)

| Gene                           | Variant                             | Genotype | Source                                                  | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                            |
|--------------------------------|-------------------------------------|----------|---------------------------------------------------------|-------------------------------------------------------|----------|-----------------------------------------------------------|
| <b>SIRT6 (centSIRT6 N308K)</b> | rs201141490<br>19:4174761 · G/A/C/T | —        | SNP not found in the data (Imputed with DR2=0.05 < 0.8) | —                                                     | weak     | centSIRT6 (N308K): rare (0.4%); does not deal with trust. |
| <b>SIRT6 (centSIRT6 A313S)</b> | rs183444295<br>19:4174748 · C/A/T   | —        | SNP not found in the data (Imputed with DR2=0.40 < 0.8) | —                                                     | weak     | centSIRT6 (A313S): rare (0.4%); does not deal with trust. |

## Losses of protective function in lipids APOC3, ANGPTL3/4/8, PCSK9 and CETP

Some rare function losses protect: an inactive copy of APOC3 (R19X, IVS2+1G>A, A43T, IVS3+1G>T) lowers the triglycerides by about 40% and the coronary disease by about 40%; ANGPTL3, ANGPTL4 (E40K) and ANGPTL8 (Q121X) with loss of function lower the triglycerides and coronary risk. PCSK9 R46L, Y142X and C679X lower the LDL throughout life (Y142X and C679X: –40%, with –88% of coronary disease in carriers). They are the targets of new drugs (PCSK9 inhibitors, ANGPTL3 and APOC3). In addition to the cured variants, all ClinVar variants were crossed with the result of loss of function or "protective" classification in these genes with the chip and imputation (table below). The CETP of lower activity (I405V, TaqIB B2, rs3764261 A) was associated with exceptional longevity in a cohort of centenarians (Barzillai 2003), with inconsistent replication; the three SNPs of CETP are a single genetic signal.

**Ageing mechanisms:** Vascular Aging; Metabolic Homeostasis decline; Chronic Inflammation

### Favourable

None.

### Unfavourable

- CETP (I405V) rs5882 AA
- CETP (TaqIB) rs708272 GG
- CETP rs3764261 CC

### Unfavourable variants (3)

| Gene                | Variant                         | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                                     |
|---------------------|---------------------------------|-----------|--------|-------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>CETP (I405V)</b> | rs5882<br>16:56982180 · G/A/C   | <b>AA</b> | chip   | A (2; 67%)                                            | weak     | CETP I405V: allele V (G) low to CETP and was more frequent in a cohort of centenarians (Barzillai 2003); inconsistent replication. |
| <b>CETP (TaqIB)</b> | rs708272<br>16:56962376 · G/A/C | <b>GG</b> | chip   | G (2; 57%)                                            | weak     | CETP TaqIB: the allele B2 (A) low to CETP and rises to HDL; same signal as rs3764261.                                              |
| <b>CETP</b>         | rs3764261<br>16:56959412 · C/A  | <b>CC</b> | chip   | C (2; 71%)                                            | weak     | CETP rs3764261: allele A low to CETP and rises the HDL (the strongest signal of HDL in GWAS).                                      |

### Non-judgmental variants (informative) (9)

| Gene                        | Variant                             | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                                                                         |
|-----------------------------|-------------------------------------|-----------|--------|-------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>APOC3 (R19X)</b>         | rs76353203<br>11:116830637 · C/G/T  | <b>CC</b> | chip   | —                                                     | strong   | APOC3 R19X: loss of function; triglycerides –40% and less coronary disease. Absent (as almost everyone).                                                               |
| <b>APOC3 (IVS2+1G&gt;A)</b> | rs138326449<br>11:116830638 · G/A   | <b>GG</b> | chip   | —                                                     | strong   | APOC3 IVS2+1G>A: loss of function (TG Exome Consortium 2014). Absent.                                                                                                  |
| <b>APOC3 (A43T)</b>         | rs147210663<br>11:116830844 · G/A/T | <b>GG</b> | chip   | —                                                     | strong   | APOC3 A43T: loss of function.                                                                                                                                          |
| <b>APOC3 (IVS3+1G&gt;T)</b> | rs140621530<br>11:116830897 · G/T   | <b>GG</b> | chip   | —                                                     | strong   | APOC3IVS3+1G>T: loss of function. Absent.                                                                                                                              |
| <b>APOC3 (-641)</b>         | rs2542052<br>11:116829268 · A/C/T   | <b>CC</b> | chip   | —                                                     | weak     | APOC3 –641: the promoter genotype associated with longevity in a cohort of centenarians (Atzmon 2006) does not have the same meaning in all studies; only informative. |
| <b>ANGPTL4 (E40K)</b>       |                                     | <b>GG</b> | chip   | —                                                     | strong   |                                                                                                                                                                        |

|                      |                                     |           |      |   |        |                                                                                                     |
|----------------------|-------------------------------------|-----------|------|---|--------|-----------------------------------------------------------------------------------------------------|
|                      | rs116843064<br>19:8364439 ·<br>G/A  |           |      |   |        | ANGPTL4 E40K: loss of function; lower triglycerides and less coronary disease (Dewey 2016). Absent. |
| <b>PCSK9 (R46L)</b>  | rs11591147<br>1:55039974 ·<br>G/A/T | <b>GG</b> | chip | — | strong | PCSK9 R46L LDL –15% and coronary artery disease –47% in patients (Cohen 2006). Absent.              |
| <b>PCSK9 (Y142X)</b> | rs67608943<br>1:55046549 ·<br>C/G/T | <b>CC</b> | chip | — | strong | PCSK9 Y142X: loss of function; LDL –40%. Absent.                                                    |
| <b>PCSK9 (C679X)</b> | rs28362286<br>1:55063542 ·<br>C/A/T | <b>CC</b> | chip | — | strong | PCSK9 C679X: loss of function; LDL –40%. Absent.                                                    |

#### SNP not found in the data (1)

| Gene                   | Variant                               | Genotype | Source                    | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                        |
|------------------------|---------------------------------------|----------|---------------------------|-------------------------------------------------------|----------|-------------------------------------------------------|
| <b>ANGPTL8 (Q121X)</b> | rs145464906<br>19:11240198 ·<br>C/G/T | —        | SNP not found in the data | —                                                     | moderate | ANGPTL8 Q121X: loss of function; lower triglycerides. |

## APOE, TOMM40/APOC1 and klotho (KL-VS)

APOE is the strongest locus of longevity in all GWAS: short  $\epsilon 4$  and long life  $\epsilon 2$ . TOMM40 and APOC1 are next to each other and their signals are largely APOE; TOMM40 polyT (rs10524523, 523) is not read in the chip. The KL-VS of the klotho (F352V, rs9536314, and C370S, rs9527025, inherited together) has an advantage in heterozygosity: 1 copy is associated with more klotho, better cognition and less effect of  $\epsilon 4$ ; 2 copies are associated with the opposite.

**Ageing mechanisms:** Protein aggregation; Neuroinflammation; Vascular Aging; Lysosome dysfunction; Endocrine dysregulation

**Favourable**

None.

### Unfavourable

None.

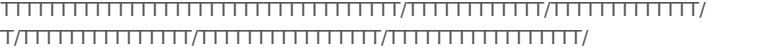
### Typical / intermediate variants (4)

| Gene             | Variant                            | Genotype  | Source                 | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                |
|------------------|------------------------------------|-----------|------------------------|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------|
| <b>APOE (ε4)</b> | rs429358<br>19:44908684<br>T/C     | <b>TT</b> | imputation<br>DR2=0.99 | C (0; 16%)                                            | strong   | APOE ε4 (C allele): the highest genetic factor of mortality and Alzheimer's.                                  |
| <b>APOE (ε2)</b> | rs7412<br>19:44908822<br>C/T       | <b>CC</b> | chip                   | C (2; 94%)                                            | strong   | APOE ε2 (T allele): the allele most associated with exceptional longevity; the genotype C/C means without ε2. |
| <b>TOMM40</b>    | rs2075650<br>19:44892362<br>A/G    | <b>AA</b> | chip                   | G (0; 13%)                                            | moderate | TOMM40 rs2075650: the G allele (in connection with ε4) was associated with less longevity (Deelen 2011).      |
| <b>TOMM40</b>    | rs12972156<br>19:44884202<br>C/A/G | <b>CC</b> | imputation<br>DR2=0.97 | G (0; 14%)                                            | weak     | TOMM40 rs12972156: G allele in connection with ε4.                                                            |

### Non-judgmental variants (informative) (3)

| Gene                    | Variant                           | Genotype  | Source                 | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                   |
|-------------------------|-----------------------------------|-----------|------------------------|-------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------|
| <b>APOC1</b>            | rs4420638<br>19:44919689<br>A/G   | <b>AG</b> | imputation<br>DR2=1.00 | —                                                     | weak     | APOC1 rs4420638: the G allele is mainly a marker of ε4; without ε4, the residual effect is small and not scored. |
| <b>KL (KL-VS F352V)</b> | rs9536314<br>13:33054001<br>T/A/G | <b>TT</b> | chip                   | —                                                     | moderate | KL-VS: the advantage is to have 1 copy (G); no copies = typical klotho.                                          |
| <b>KL (KL-VS C370S)</b> | rs9527025<br>13:33054056<br>G/C/T | <b>GG</b> | imputation<br>DR2=1.00 | —                                                     | moderate | KL-VS C370S (C allele): inherited with F352V; confirms the reading.                                              |

### SNP not found in the data (1)

| Gene                      | Variant                                                                                                          | Genotype | Source                    | Unfavourable allele (copying; frequency in reference) |
|---------------------------|------------------------------------------------------------------------------------------------------------------|----------|---------------------------|-------------------------------------------------------|
| <b>TOMM40 (polyT 523)</b> | rs10524523<br>19:44899792 ·  | —        | SNP not found in the data | —                                                     |



## CDKN2A/B (9p21): senescence and life expectancy

9p21 (CDKN2B-AS1/ANRIL, next to the senescence genes p16INK4a and p15INK4b) is, after APOE, Lp(a) and tobacco (CHRNA3/5), the locus that shortens parental life expectancy in GWAS (Pilling 2017, Timmers 2019), especially through coronary disease. The variants are in the same block and count as a single signal: the risk haplotype has 5 SNPs (4 in the chip and 1 imputed;  $r$  0.92–0.998 connection in the reference population). Four are in the table below; the fifth, rs4977574 G/G (chip), is in the Advanced Cardiovascular report. Rs2811712 of CDKN2B was associated with physical function in the elderly (Melzer 2007).

**Ageing mechanisms:** Cell Senescence; Vascular Aging; Exhaustion of Staminary Cells

### Favourable

None.

### Unfavourable

- CDKN2B-AS1 (9p21) rs1333049 CC
- CDKN2B-AS1 (9p21) rs10757278 GG
- CDKN2B-AS1 (9p21) rs1333042 GG
- CDKN2B-AS1 (9p21) rs1556516 CC

### Unfavourable variants (4)

| Gene                     | Variant                               | Genotype  | Source                 | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                                         |
|--------------------------|---------------------------------------|-----------|------------------------|-------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------------------------------------|
| <b>CDKN2B-AS1 (9p21)</b> | rs1333049<br>9:22125504 ·<br>G/A/C    | <b>CC</b> | chip                   | C (2; 47%)                                            | strong   | 9p21 rs1333049: the allele C is the coronary risk (the strongest coronary sign of GWAS).                                               |
| <b>CDKN2B-AS1 (9p21)</b> | rs10757278<br>9:22124478 ·<br>A/C/G/T | <b>GG</b> | chip                   | G (2; 47%)                                            | strong   | 9p21 rs10757278: the allele G is that of coronary risk; same block.                                                                    |
| <b>CDKN2B-AS1 (9p21)</b> | rs1333042<br>9:22103814 ·<br>A/C/G    | <b>GG</b> | chip                   | G (2; 50%)                                            | moderate | 9p21 rs1333042: allele A was associated with more longevity of the father (Wright 2019); same block.                                   |
| <b>CDKN2B-AS1 (9p21)</b> | rs1556516<br>9:22100177 ·<br>G/A/C/T  | <b>CC</b> | imputation<br>DR2=1.00 | C (2; 50%)                                            | moderate | 9p21 rs1556516: parental life expectancy locus (Pilling 2017); the C allele follows the coronary risk of rs1333049 C (same haplotype). |

### Typical / intermediate variants (1)

| Gene          | Variant                              | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                        |
|---------------|--------------------------------------|-----------|--------|-------------------------------------------------------|----------|-------------------------------------------------------------------------------------------------------|
| <b>CDKN2B</b> | rs2811712<br>9:21998036 ·<br>G/A/C/T | <b>AA</b> | chip   | A (2; 90%)                                            | weak     | CDKN2B rs2811712: G allele was associated with less physical limitation in the elderly (Melzer 2007). |

# Life expectancy of GWAS and longevity scores

The GWAS of life expectancy of parents (about 1 million people) found a dozen loci outside the APOE, almost all linked to cardiovascular disease, tobacco or immunity. Here we read those that have clear sense and four scores of the PGS Catalog: longevity (330 variants); Tesi 2020), exceptional longevity (4 variants; Deelen 2019, age of death of the father (398 variants) and male mortality (4 092 variants). The four scores had already been calculated in the Exceptional Longevity report (profile §45) and were recalculated here with a new index: the values coincide. A locus of parental life expectancy is joined.

**Ageing mechanisms:** Vascular Aging; Chronic Inflammation; Immunosenescence; Metabolic Homeostasis Decline

## Polygenic and loci scores

| Score                                                | Result                        | Percentage (reference population) | Reading and method                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------------------------------------------------------|-------------------------------|-----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Longevity (330 variants)</b>                      | <b>Favourable</b>             | the 92nd percentile · z=1.48      | Higher score = more likely to reach very advanced ages. PGS000906 (PGS Catalog; Clumping and Thresholding (C+T); Longevity; PGP000237, PMID 33216869). Variants with genotype: 312 of 330 (covering of 90.1% coefficients).                                                                                                                                                                                                                 |
| <b>Exceptional longevity (4 variants)</b>            | <b>Typical / intermediate</b> | the 47th percentile · z=0.29      | Only 4 variants (APOE, FOXO3 and others); very low power. PGS002795 (PGS Catalog; Genome-wide signifier SNPs; Longevity (>90th survival percentile); PGP000378, PMID 34091660). Variants with genotype: 4 of 4 (covering of 100.0% coefficients).                                                                                                                                                                                           |
| <b>Age of death of the father (398 variants)</b>     | <b>Typical / intermediate</b> | the 30th percentile · z=-0.56     | Higher score = father with later death, on average. PGS001393 (PGS Catalog; snpnet; Father's acts at death; PGP000244, PMID 35324888). Variants with genotype: 328 of 398 (covering of 80.4% coefficients).                                                                                                                                                                                                                                 |
| <b>Mortality by all causes, men (4 092 variants)</b> | <b>Typical / intermediate</b> | the 61st percentile · z=0.26      | Higher score : more mortality. PGS000319 (PGS Catalog; Composite PRS (component scores combined by log(HR)); All-cause mortality (male); PGP000095, PMID 32758451). Variants with genotype: 3716 of 4092 (covering of 88.7% coefficients).                                                                                                                                                                                                  |
| <b>Parental life expectancy (GWAS loci)</b>          | <b>Favourable</b>             | the 16th percentile · z=-1.13     | Higher score = more alleles that shorten the life expectancy of parents. Loci score of GWAS Catalog (p≤510-8; 1 variant by 500 kb window; simple count of risk alleles, without weighting, because the studies use different scales: 13 of 14 loci with genotype. Studies: Pilling LC (Parental longevity (combined parentage at death)); Timers PR (Parental lifespan). Variants with genotype: 13 of 14 (covering of 92.9% coefficients). |

### Favourable

- Longevity (330 variants) (the 92nd percentile)
- Parental life expectancy (GWAS loci) (the 16th percentile)
- HLA-DQA1 rs34831921 AC

### Unfavourable

None.

### Favourable variants (1)

| Gene            | Variant                       | Genotype | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                                                                |
|-----------------|-------------------------------|----------|--------|-------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>HLA-DQA1</b> | rs34831921 · 6:32622991 · C/A | AC       | chip   | C (1; 90%)                                            | weak     | HLA-DQA1/DRB1: locus of parental life expectancy (autoimmunity and infections). Meaning: GWAS Catalog: Parental lifespan (Joshi PK 2017; p=4e-08); allele A ↑ |

### Typical / intermediate variants (3)

| Gene                   | Variant                         | Genotype | Source                | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                           |
|------------------------|---------------------------------|----------|-----------------------|-------------------------------------------------------|----------|----------------------------------------------------------------------------------------------------------|
| <b>CHRNA3/5 (HYKK)</b> | rs8042849 · 15:78525587 · C/A/T | TC       | imputation · DR2=1.00 | C (1; 39%)                                            | moderate | CHRNA3/5: the T allele is associated with more years of life of parents (Timmers 2019), through tobacco; |

|                      |                                   |           |                        |            |          |                                                                                                                                   |
|----------------------|-----------------------------------|-----------|------------------------|------------|----------|-----------------------------------------------------------------------------------------------------------------------------------|
|                      |                                   |           |                        |            |          | without tobacco, the effect almost disappears.                                                                                    |
| <b>FURIN</b>         | rs6224<br>15:90880313<br>· G/C/T  | <b>GT</b> | imputation<br>DR2=0.99 | T (1; 47%) | moderate | FURIN rs6224: the G allele is associated with more years of parents' life (Timmers 2019); effect via lipids and coronary disease. |
| <b>5q33.3 (EBF1)</b> | rs2149954<br>5:158393594<br>· C/T | <b>TC</b> | chip                   | C (1; 65%) | weak     | 5q33.3 rs2149954: the T allele was associated with surviving the 90 years (Deelen 2014).                                          |

### Non-judgmental variants (informative) (1)

| Gene                   | Variant                            | Genotype  | Source | Unfavourable allele (copying; frequency in reference) | Evidence | Interpretation                                                                                                         |
|------------------------|------------------------------------|-----------|--------|-------------------------------------------------------|----------|------------------------------------------------------------------------------------------------------------------------|
| <b>IL6 (rs2069837)</b> | rs2069837<br>7:22728408<br>· A/C/G | <b>AA</b> | chip   | —                                                     | weak     | IL6 rs2069837: associated with longevity in a cohort of centenaries (Zeng 2016); without replication in other cohorts. |

## 84 map of aging mechanisms

Genetic status of each 84 mechanism of the target list, from all reports integrated in the profile. **15** favourable, **10** unfavourable, **21** mixed, 9 typical, **28** without possible genetic reading with chip and imputation, and 1 outside the scope by rule. Many mechanisms of the nucleus, RNA and peroxisomes do not have common germ variants with known effect: their state is measured in blood or tissues (epigenetic clocks, somatic sequencing), not in the inherited DNA. The sections indicated are those of the genetic profile.

| Mechanism                                     | Genetic status            | Main findings                                                                                                                                                                                                             | Profile                              |
|-----------------------------------------------|---------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------|
| <b>Main brands</b>                            |                           |                                                                                                                                                                                                                           |                                      |
| <b>Genomic instability</b>                    | <b>Favourable</b>         | Common routes of repair without alleles of lower efficiency 0 variants P/LP plausible in 120 genes of hereditary disease; low predicted Y loss (dose in the 86th percentile). TP53 Arg72/Arg72 is a compromise.           | \$36, \$81, \$65                     |
| <b>DNA Damage and DNA Repair Deficit</b>      | <b>Mixed</b>              | XRCC1, XRCC3, ERCC2 and XPC with common genotypes; but NAT2 slow acetylator, probable zero GSTM1 and GSTA1*B/*B pass through more fertilizers (aromatic amines, aldehydes); ATM D1853N A/A (radiation).                   | \$35, \$36, \$93, \$94               |
| <b>Hayflick's Telomeres and Limit</b>         | <b>Unfavourable</b>       | Genetically shorter telomeres (87 loci in the 25th percentile; 7 SNPs in 24; TERT rs10069690 T/T. TERT rs2736100 C/C elongates, but with more glioma and clonal expansions (commitment)).                                 | \$37, \$24                           |
| <b>Epigenetic Changes</b>                     | <b>Typical</b>            | Intrinsic clock (IEAA) in the 48th percentile, GrimAge in the 7 (indicative), PhenoAge in the 78 and Hannum in the 68; MTHFR compound heterozygous and genetic homocysteine in the 92nd percentile (methylation).         | \$64, \$13                           |
| <b>Proteostasis loss</b>                      | <b>Mixed</b>              | Favourable neuron (APOE ε3/ε3, MAPT H2/H2, without GBA1 or LRRK2 G2019S); unfavourable liver (alfa-1 antitrypsin Z protein folds poorly and accumulates).                                                                 | \$39, \$25, \$77                     |
| <b>Autophagy / Deregulated Macroautophagy</b> | <b>Favourable</b>         | FOXO3 homozygous for the 4 alleles of longevity (active to autophagy); ATG16L1 T300A and MTOR rs2295080 heterozygotes (on average).                                                                                       | \$38, \$45, this report              |
| <b>Nutrient Sensing Deregulation</b>          | <b>Favourable</b>         | FOXO3 homozygous longevity; IGF-1 genetic in the 7th percentile; GHR d3 with 1 copy (rs6873545 T/C); IGF1R rs2229765 G/A. BCAT2 C/C (BCAA a little higher, small effect).                                                 | \$38, \$80, this report              |
| <b>Mitochondrial dysfunction</b>              | <b>Favourable</b>         | PGC-1α Gly482Gly, normal NDUF6, β-oxidation without pathogenic variants, mtDNA HV without classical variants. Small: SIRT3 V208I and UCP1 –3826 heterozygotes.                                                            | \$17, \$72                           |
| <b>Cell Senescence</b>                        | <b>Unfavourable</b>       | 9p21 (CDKN2A/B) homozygous for the risk haplotype in 5 SNPs (4 on chip, 1 imputed); CDKN1A (p21) G/A; GDF-15 genetic in the 83rd percentile; IL6 –174 G/G (plus SASP). TP53 Arg72 favors apoptosis instead of senescence. | \$26, \$37, \$97, \$101, this report |
| <b>Exhaustion of Staminary Cells</b>          | <b>Mixed</b>              | Telomeric reserve slightly below the average (the 25th percentile), but without JAK2 46/1, SH2B3 C/C and CHIP score in the 30th percentile.                                                                               | \$40, \$57                           |
| <b>Intercellular Communication</b>            | <b>Typical</b>            | Favourable hormones (genetic testosterone in the 89th percentile, low IGF-1) with high SHBG (the 87th percentile); typical cytokines.                                                                                     | \$43, \$65, \$38                     |
| <b>Chronic Inflammation</b>                   | <b>Mixed</b>              | Low genetic CRP (the 12th percentile), IL6R Asp358Ala (1 copy), TNF –308 G/G and HLA autoimmune clean; against GlycA in the 90th percentile, IL6 –174 G/G and gingivitis in 90.                                           | \$44, \$97, \$18, \$88               |
| <b>Dysbiosis</b>                              | <b>Favourable</b>         | FUT2 secretor, persistent lactase, inflammatory bowel disease in the 2-9 percentiles, celiac in the 4, NOD2 without variants; CARD9 S12N T/T (antifungal immunity) and normal FMO3 (TMAO).                                | \$42, \$84                           |
| <b>Epigenetics, nucleus and genome</b>        |                           |                                                                                                                                                                                                                           |                                      |
| <b>Epigenetic Derivative</b>                  | <b>Typical</b>            | No known causal gene; the loci of epigenetic clocks give a typical intrinsic clock (the 48th percentile).                                                                                                                 | \$64                                 |
| <b>Loss of Epigenetic Fidelity</b>            | <b>No genetic reading</b> | No germinal variants with known effect are measured with epigenetic clocks in the blood, not with the chip.                                                                                                               | —                                    |

|                                                                                     |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |
|-------------------------------------------------------------------------------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| <b>Loss of Heterochromatin</b>                                                      | No genetic reading | Without direct reading; SIRT6 of the centenaries (centSIRT6) is not read with confidence.                                                                                                                                                                                                                                                                                                                                                                                                                                                                | This report             |
| <b>Epigenetic Memory Deregulation</b>                                               | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Nucleosome and Histone alterations</b>                                           | No genetic reading | No useful genetic reading (rare variants of histones and their modifiers are not on the chip).                                                                                                                                                                                                                                                                                                                                                                                                                                                           | —                       |
| <b>Aberrant Transcriptional Reprogramming</b>                                       | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Disorganization of Nuclear Architecture</b>                                      | No genetic reading | Without genetic reading; LMNA has no plausible P/LP variants (the only reading was probe artifact).                                                                                                                                                                                                                                                                                                                                                                                                                                                      | \$81                    |
| <b>Deterioration of the Nuclear Blade and the Domains Associated with the Blade</b> | No genetic reading | Without genetic reading (LMNA without P/LP; centSIRT6, which reinforces the binding to lamina A, is not read).                                                                                                                                                                                                                                                                                                                                                                                                                                           | \$81, this report       |
| <b>Retrotransposing activation</b>                                                  | No genetic reading | Without genetic reading; centSIRT6 (which stops LINE-1) is not read.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | This report             |
| <b>Endogenous Retrovirus dysregulation</b>                                          | No genetic reading | The chip does not read the endogenous retrovirus insertions.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | —                       |
| <b>Somatic Mutation Accumulation</b>                                                | Mixed              | Common favourable repair and protective pigmentation (in childhood in the 1st percentile, low skin cancers); but weak II phase detoxification (Slow acetylator NAT2, GSTA1*B/*B and GCLM with a low glutathione synthesis allele), TERT axis pulling up (glioma in the 91st percentile) and colorectal cancer above the median in the complete scores (94.3 and 78.5 percentiles; early onset in 82.3; polyps in the 92.7 loci). Together, more somatic mutations predicted in the colon than in the skin: the actionable point is colorectal screening. | \$24, \$36, \$94, \$102 |
| <b>Somatic / Tecdual Mosaicism</b>                                                  | Favourable         | Y loss in the 37th percentile and Y dosage in the 86 (less predicted loss).                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | \$36, \$65              |
| <b>Clonal hematopoiesis (CHIP)</b>                                                  | Typical            | Typical to low predisposition: without JAK2 46/1, SH2B3 C/C, CHIP score in the 30th percentile; TCL1A G/G (without the protective allele) and TERT rs7705526 heterozygous. The real state (acquired clones) is not seen in the chip.                                                                                                                                                                                                                                                                                                                     | \$40, \$57, \$64        |
| <b>Dysfunction of Primary Cyls</b>                                                  | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>RNA and proteostasis</b>                                                         |                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                         |
| <b>RNA dysregulation</b>                                                            | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Critical Transcription</b>                                                       | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Splicing Errors</b>                                                              | No genetic reading | Without genetic reading (the rare splicing variants are almost not on the chip).                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | —                       |
| <b>Increased Transcriptional Elongation Velocity</b>                                | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Transcriptional Decline Depending on Gene Size</b>                               | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Protein Aggregation</b>                                                          | Mixed              | Favourable brain: APOE ε3/ε3, MAPT H2/H2, TTR V122I absent, Parkinson in the 8th percentile; Alzheimer's without APOE of 39 variants in 91 (genomic no 52). Liver: protein Z.                                                                                                                                                                                                                                                                                                                                                                            | \$25, \$39, \$34        |
| <b>Molecular Platerone Failure</b>                                                  | No genetic reading | No direct genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | —                       |
| <b>Proteasome dysfunction</b>                                                       | No genetic reading | No genetic reading.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | —                       |
| <b>Lysosome dysfunction</b>                                                         | Mixed              | GBA1 and LRRK2 G2019S absent and Parkinson low; but LRRK2 rs76904798 T/T, TMEM175 and GPNMB with risk alleles.                                                                                                                                                                                                                                                                                                                                                                                                                                           | \$25, \$39              |

|                                                                |                    |                                                                                                                                                                                |                  |
|----------------------------------------------------------------|--------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| <b>Autophagy Mediated by Reduced Chaperones</b>                | No genetic reading | No direct genetic reading (LRRK2 G2019S absent).                                                                                                                               | \$25             |
| <b>ER-fagia Defective</b>                                      | No genetic reading | Without genetic reading, it is one of the pathways that eliminate Z protein in the liver.                                                                                      | —                |
| <b>Remodelling of the Endoplasmic Reticulum</b>                | No genetic reading | No genetic reading.                                                                                                                                                            | —                |
| <b>Endoplasmic Reticulum Disorder</b>                          | Unfavourable       | The Z protein (PiMZ) accumulates in the reticulum of the hepatocyte; PERKrs7571971C/C (favourable) does not compensate.                                                        | \$39, \$51, \$77 |
| <b>Mythochondria, redox and peroxisomes</b>                    |                    |                                                                                                                                                                                |                  |
| <b>Oxidative Phosphorylation Decline</b>                       | Favourable         | PGC-1α Gly482Gly, NDUFB6 and COQ2 without risk variants, mtDNA without classical variants.                                                                                     | \$17             |
| <b>Changes in Mitochondrial Dynamics</b>                       | No genetic reading | No useful genetic reading.                                                                                                                                                     | —                |
| <b>Mitochondrial Defective Quality Control</b>                 | Favourable         | FOXO3 homozygous longevity (mythophagy); no variants read at risk in the pathways of mitophagy.                                                                                | \$17, \$38       |
| <b>Mythochondrial Calcium Deregulation</b>                     | No genetic reading | No genetic reading.                                                                                                                                                            | —                |
| <b>Mutations in the DNA Mitochondrial</b>                      | Favourable         | Inherited: haplogroup HV without m.3243A>G, m.8344A>G, m.11778G>A or m.14484T>C. Somatic mutations and heteroplasma are not seen.                                              | \$11, \$19       |
| <b>Release of Mythochondrial DAMPs</b>                         | Typical            | HMGB1 (loci) in the 47th percentile; without direct reading.                                                                                                                   | \$97             |
| <b>47 mechanism of the list (mitochondrial redox cofactor)</b> | Out of scope       | Out of the scope of these reports, by rule of the genetic plan.                                                                                                                | —                |
| <b>Redox imbalance</b>                                         | Mixed              | NRF2 –617C/C, NQO1 *1/*1, GCLC and SOD2 favourable; Probable null GSTM1, GSTA1*B/*B, low expression PON1, MPO –463G/G and GCLM heterozygote.                                   | \$16, \$35, \$93 |
| <b>Oxidative stress</b>                                        | Mixed              | The same pattern: strong antioxidant regulation, weak II phase conjugation.                                                                                                    | \$16, \$35       |
| <b>Peroxisomal dysfunction</b>                                 | No genetic reading | No genetic reading.                                                                                                                                                            | —                |
| <b>Failure in Peroxisome-Mythochondria Communication</b>       | No genetic reading | No genetic reading.                                                                                                                                                            | —                |
| <b>Metabolism and endocrine</b>                                |                    |                                                                                                                                                                                |                  |
| <b>Metabolic Homeostasis Decline</b>                           | Mixed              | HbA1c genetics in the 7th percentile and fasting glucose in 9.5; 2 polygenic diabetes in the 57-69, TCF7L2 rs7903146 C/T, SLC30A8 and KCNJ11 percentiles in risk homozygosity. | \$52             |
| <b>Insulin resistance</b>                                      | Mixed              | Fasting insulin in the 5th percentile (12 variants) and gluteal-femoral fat in 95; against trunk fat in 98, FTO, KLF14 and IRS1 heterozygotes.                                 | \$52, \$98       |
| <b>Hyperglycemia and Intolerantness to Glucose</b>             | Favourable         | HbA1c and genetically low fasting glucose; MTNR1B C/G (late meals are worse).                                                                                                  | \$52             |
| <b>Dysfunction of adipose tissue</b>                           | Mixed              | Boundary fat in the 98th percentile and abdominal subcutaneous fat in 87; against gluteal-femoral fat in 95, adiponectin (loci) in 89 and FABP4 in 18.                         | \$98, \$59       |
| <b>Thermogenesis decline</b>                                   | Typical            | UCP1 Ala64, ADRB3 Trp64 and DIO2 Thr92 favourable; UCP1 –3826 heterozygous and TBX15-WARS2 T/T.                                                                                | \$59, \$86       |
| <b>Endocrine dysregulation</b>                                 | Favourable         | Genetic testosterone in the 89th percentile, prolactin in the 12, typical thyroid High SHBG (the 87th percentile).                                                             | \$43, \$65, \$86 |
| <b>Decline of the Estrogen Signal</b>                          | Typical            | Little relevant in a man; ESR1 with common alleles                                                                                                                             | \$43             |
| <b>Thyroid alterations</b>                                     | Typical            | Hypothyroidism and hyperthyroidism in the 34 and 38 percentiles; Low genetic TSH and high free T4 (without judgment).                                                          | \$86             |

|                                                                 |                           |                                                                                                                                                                                                                                                                                                                 |                                             |
|-----------------------------------------------------------------|---------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Accumulation of Advanced Glycation Final Products (AGEs)</b> | <b>Favourable</b>         | HbA1c genetics in the 7th percentile: less baseline glycation.                                                                                                                                                                                                                                                  | \$28, \$52                                  |
| <b>Immunity, barriers, matrix and vessels</b>                   |                           |                                                                                                                                                                                                                                                                                                                 |                                             |
| <b>Inflammaging</b>                                             | <b>Mixed</b>              | CRP low and IL6R Ala (favourable) against GlycA in the 90th percentile and IL6 –174 G/G.                                                                                                                                                                                                                        | \$44, \$97                                  |
| <b>Immunosenescence</b>                                         | <b>Mixed</b>              | Good innate defense (MBL2 A/O without deficiency, TLR4 Asp299Gly absent, COVID-19 severe in percentile 4.7, sepsis in 3–23, neutrophils in 20); against polygenic tendency for bacterial mucosal infections (pneumonia in 84, urinary infection in 94, otitis in 85; zone in 75) and FCGR2A H131R heterozygous. | \$44, \$103                                 |
| <b>Thymic Involution</b>                                        | <b>No genetic reading</b> | No direct genetic reading (without score or variants with known effect on thymus).                                                                                                                                                                                                                              | \$103                                       |
| <b>Loss of T Naive Cells</b>                                    | <b>No genetic reading</b> | Without direct genetic reading; the Pneumococcus and Immunosenescence report only conditionally binds to the weaker response to vaccines with age.                                                                                                                                                              | \$103                                       |
| <b>Repertory TCR restriction</b>                                | <b>No genetic reading</b> | Without genetic reading (the repertoire is formed throughout life); HLA of class II typed only by markers.                                                                                                                                                                                                      | \$18                                        |
| <b>Intestinal Barreira dysfunction</b>                          | <b>Favourable</b>         | Very low inflammatory bowel and celiac disease, secretory FUT2; CARD9 T/T and ATG16L1 T300A heterozygous.                                                                                                                                                                                                       | \$42                                        |
| <b>Remodelling of the extracellular matrix</b>                  | <b>Mixed</b>              | Discal disc and hernia very low (3–17 percentiles), COL3A1 and GDF5 favourable; COL5A1 rs12722 T/T, MMP9 and LOXL1 with weak risk alleles.                                                                                                                                                                      | \$28, \$50, \$74                            |
| <b>Stiffness of the Extracellular Matrix</b>                    | <b>Favourable</b>         | Arterial stiffness score (pulse wave velocity) in the 12th percentile.                                                                                                                                                                                                                                          | \$28                                        |
| <b>Fragmentation of Elastin</b>                                 | <b>Mixed</b>              | Genetically wider ascending aorta (the 79th percentile; FBN1 and ULK4 with larger diameter alleles ELN and LOXL1 mixed).                                                                                                                                                                                        | \$87, \$28                                  |
| <b>Vascular Aging</b>                                           | <b>Mixed</b>              | Polygenic strengths (coronary disease in the 5th percentile, atrial fibrillation no 2, stroke no 14-19, dilated cardiomyopathy no 2) against focal weaknesses (9p21 homozygous, white matter lesions no 92, venous thrombosis no 86, aortic stenosis no 95, aorta no 79, venous thromboembolism no 95.1).       | \$26, \$49, \$87, \$95, \$100, \$101, \$105 |
| <b>Arterial stiffness</b>                                       | <b>Favourable</b>         | Arterial stiffness score in the 12th percentile.                                                                                                                                                                                                                                                                | \$28                                        |
| <b>Rhythms, sleep and brain</b>                                 |                           |                                                                                                                                                                                                                                                                                                                 |                                             |
| <b>Circadian disorders</b>                                      | <b>Mixed</b>              | Morning Chronotype (the 86th percentile) and daytime sleepiness in 1 MTNR1B C/G and CLOCK G/A.                                                                                                                                                                                                                  | \$41                                        |
| <b>Desynchronization of Biological Clocks</b>                   | <b>Typical</b>            | No clear signal beyond MTNR1B (late meals).                                                                                                                                                                                                                                                                     | \$41, \$52                                  |
| <b>Sleep disturbance</b>                                        | <b>Unfavourable</b>       | Insomnia (129 loci) in the 89th percentile, duration of genomic sleep in 17 and restless legs in 77.                                                                                                                                                                                                            | \$41, \$67, \$70                            |
| <b>Glymphatic dysfunction</b>                                   | <b>Unfavourable</b>       | Perivascular spaces: SLC13A3 C/C (2 risk alleles, the strongest locus) and LPAR1 G/A; white matter lesions in the 92nd percentile.                                                                                                                                                                              | \$100, \$25                                 |
| <b>Neuroinflammation</b>                                        | <b>Mixed</b>              | APOE ε3/ε3 and TREM2 without R47H/R62H; against microglial Alzheimer's loci (MS4A6A, EPHA1, PTK2B; 39 variant score in the 91st percentile) and central auditory pathway with a tendency to tinnitus (89.7 score, 94.6 gravity, 99.8 locus).                                                                    | \$25, \$34, \$104                           |
| <b>Hematoencephalic Barreira dysfunction</b>                    | <b>Unfavourable</b>       | Small cerebral vessels: white matter in the 92nd percentile, PMF1 C/C and COL4A2 A/C; APOE ε3/ε3 attenuates.                                                                                                                                                                                                    | \$100, \$25                                 |
| <b>Muscle, bone and frailty</b>                                 |                           |                                                                                                                                                                                                                                                                                                                 |                                             |
| <b>Sarcopenia</b>                                               | <b>Unfavourable</b>       | Appendicular lean mass in the 10th percentile and fat-free mass in the 27–30; sarcopenia "more likely" in an external report.                                                                                                                                                                                   | \$27, \$72                                  |
| <b>Thin Mass Loss</b>                                           | <b>Unfavourable</b>       | The same sign of low lean mass; low genetic physical activity (7–14 percentiles).                                                                                                                                                                                                                               | \$27, \$72, \$73                            |
| <b>Loss of muscle strength</b>                                  | <b>Mixed</b>              | Typical genomic grip strength (51–52 percentiles), but load of force loci in 2.4.                                                                                                                                                                                                                               | \$72, \$27                                  |
| <b>Muscular atrophy</b>                                         | <b>Unfavourable</b>       | Low genetic lean mass; MSTN and ACTN3 typical.                                                                                                                                                                                                                                                                  | \$27, \$72                                  |

|                                 |                    |                                                                                                                                                                 |                  |
|---------------------------------|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Neuromuscular Junction Disorder | No genetic reading | No genetic reading.                                                                                                                                             | —                |
| Osteoporosis                    | Mixed              | Fractures in the 84, ESR1, RSPO3 and SMAD3 percentile of risk; against ZBTB40, AXIN1, WLS and COL1A1 favourable and high genetic vitamin D (84–91 percentiles). | \$27, \$54       |
| Fragility                       | Unfavourable       | Fragility in the percentiles 75 (943 thousand variants) and 82 (external report); cognitive fragility in 77.                                                    | \$64, \$27, \$96 |

# The 15 unfavourable genetic points with the greatest impact

The unfavourable genetic points with the greatest probable impact on longevity, ordered by the combination of evidence, effect size and possibility of acting. Order is a clinical judgment on genetic data, not a calculation.

| # | Point                                                          | Genetics                                                                                                                                                                                                                                                                           | Evidence | Why does it matter                                                                                                                                                                                                                          | Main suggestion                                                                                                                                                                                                                    |
|---|----------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>9p21 (CDKN2A/B) risk homozygous</b>                         | 5 SNPs of the same haplotype: rs1333049 C/C, rs10757278 G/G, rs4977574 G/G and rs1333042 G/G (chip) and rs1556516 C/C (imputed, DR2 1), with r-binding 0.92–0.998 in the reference population; also abdominal aortic aneurysm and peripheral arterial disease.                     | Strong   | The common locus that shortens parental life expectancy outside of APOE, Lp(a) and tobacco (~1.3× per allele in coronary disease); acts by the senescence of smooth muscle cells, regardless of LDL.                                        | LDL/ApoB as low as possible throughout life and keep zero tobacco; discuss with the doctor a coronary calcium score to calibrate the intensity. <i>suggestion (not applied)</i>                                                    |
| 2 | <b>Alfa-1 antitrypsin PiMZ + PNPLA3 I148M (liver and lung)</b> | SERPINA1 Pi*Z C/T, PNPLA3 I148M C/G, MBOAT7 and GCKR P446L with 1 allele, HFE H63D; high genetic enzymes (ALT in the 87th percentile, GGT in the 94); emphysema (42 variants) in 96.2, the most likely pathway of secondary pulmonary hypertension (only with tobacco or tobacco). | Strong   | In patients with liver fat, advanced fibrosis and cirrhosis are ~2–3× more frequent; the PNPLA3 is added. In the lung, Z only has tobacco.                                                                                                  | Zero alcohol, no smoke or vapors, low hepatic fat; periodic liver elastography and ALT monitored with isotretinoin and rapamycin; ferritin < 100 without supplementary iron. <i>suggestion (not applied)</i>                       |
| 3 | <b>Mental health: depression, ADHD and insomnia</b>            | Major the 95th percentile depression, ADHD in 94 and 97, insomnia (loci) in 89, bipolar 79 disease and schizophrenia in 88.                                                                                                                                                        | Moderate | Depression and ADHD are associated with fewer years of life (accidents, substances, less adherence to care) and add up to insomnia; the genetic load is high, but the environment and treatment count more.                                 | Regular monitoring of mood and sleep with the physician (low dose lithium is already in the plane); cognitive-behavioral therapy for insomnia, daily exercise and melatonin 0.3–1 mg. <i>suggestion (not applied)</i>              |
| 4 | <b>Venous thrombosis, pulmonary embolism and fibrinogen</b>    | Venous thrombosis in the 86 and 84 percentiles and loci load in 95.1; pulmonary embolism in the 90.8 loci (in a genomic score in 1.4); fibrinogen in 92 F11 rs2289252 T/T, FGG and PROCOR with risk alleles (without factor V Leiden or prothrombin; group O).                     | Moderate | Venous thromboembolism is a preventable cause of death in surgery, immobilization and long trips, and the thromboembolic pathway is the most plausible form of pulmonary hypertension for this profile (BMP2 is not evaluated by the chip). | Mechanical and pharmacological prophylaxis in surgeries and hospitalizations, early mobilization, sock and hydration in long trips; knowing the signs (swollen and painful legs, sudden air loss). <i>suggestion (not applied)</i> |
| 5 | <b>Aortic aorta and valve</b>                                  | Diameter of the ascending aorta in the 79th percentile (FBN1 and ULK4), aortic stenosis (492 loci) in the 95, TEX41 and ALPL homozygous; 9p21 risk for abdominal aneurysm.                                                                                                         | Moderate | Aortic dilation and aortic stenosis progress without symptoms for years; detected early, they are treated in time.                                                                                                                          | Base echocardiogram and repeated periodically, to be discussed with the doctor; avoid fluoroquinolones when there is an alternative (associated with aneurysm and dissection of the aorta). <i>suggestion (not applied)</i>        |
| 6 | <b>Thin mass and low force (sarcopenia)</b>                    | Appendicular lean mass in the 10th percentile, fat-free mass in the 27–30, force loci load in the 2.4, fragility in the 75–82 and inclination to physical activity in the 7–14.                                                                                                    | Moderate | Little muscle mass and little strength are among the strongest predictors of mortality and dependence on old age; with retatrutide, lean mass loss is an additional risk.                                                                   | Strength training 3 times a week with progression, protein ~140 g/day in 3–4 meals and creatine (as in the plane); DEXA to follow the lean mass. <i>suggestion (not applied)</i>                                                   |
| 7 | <b>Small cerebral vessels and Alzheimer without APOE</b>       | White matter lesions in the 92nd percentile, PMF1 C/C, SLC13A3 C/C (perivascular spaces); Alzheimer without APOE (39 variants) in 91 and mild cognitive impairment in 98.6 (low reliability); BDNF Val66Met.                                                                       | Moderate | Small vessel disease is the main vascular cause of cognitive decline; the absence of ε4 maintains the absolute risk of Alzheimer's disease.                                                                                                 | Regular aerobic exercise, sufficient sleep, low LDL and glucose and omega-3 at the dose of the plan (3.2–4 g/day); baseline cognitive assessment, to be discussed. <i>suggestion (not applied)</i>                                 |

|                                                   |                                                                                                                                                                                                                                                                         |                 |                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>8 Colorectal cancer and polyyps (polygen)</b>  | Complete scores in the 94.3 (95 variants) and 78.5 (1 million variants); early onset (< 50 years) in 82.3; load of polyyps loci and adenomas in 92.7 and colorectal cancer in 83.1 BMP4 in homozygosity, 8q24, SMAD7, COLCA2. Without APC I1307K, MUTYH or CHEK2 I157T. | Moderate        | Colorectal cancer is one of the most frequent and almost avoidable cancers with colonoscopy (polyyps remove before malignancy). The small scores of the Cancer Risk report (typical) underestimated this risk.                                         | First quality colonoscopy at the age of 45 (or already, if this age has passed), in a center with good rate of detection of adenomas, more research of occult blood in feces by immunochemical test (FIT) in intervals; next interval defined by the findings; earlier if there is family history. This reading replaces that of the Cancer Risk report (\$24), in which colorectal genetics did not justify anticipating screening. <i>suggestion (not applied)</i> |
| <b>9 Weak II phase detoxification (slow NAT2)</b> | NAT2 slow acetylator (*5/*5), probable null GSTM1, GSTA1*B/*B, low expression PON1; bladder cancer loci scores in the 86-95 percentiles (the genomic is typical).                                                                                                       | Moderate        | Aromatic and heterocyclic amines (smoker, charred meat, hair dyes, solvents) spend more time in the body and increase bladder cancer.                                                                                                                  | Avoid carbonized and smoked meat, smoking and hair dyes with aromatic amines; good hydration; routine urine analysis (blood in the urine) to be discussed with the doctor. <i>suggestion (not applied)</i>                                                                                                                                                                                                                                                           |
| <b>10 Central Adiposity and Loci of Diabetes</b>  | Boundary fat in the 98th percentile, subcutaneous abdominal in 87; TCF7L2 rs7903146 C/T, SLC30A8 and KCNJ11 in homozygosity, 2 type diabetes in the 57-69 percentiles.                                                                                                  | Moderate        | Central fat is the metabolic phenotype most linked to cardiovascular and hepatic disease; low genetic glucose compensates in part.                                                                                                                     | Maintain the current plan (retathrutide and dapagliflozin) with strength training; early main meals (MTNR1B); follow visceral fat by DEXA. <i>suggestion (not applied)</i>                                                                                                                                                                                                                                                                                           |
| <b>11 Short telomeres and TERT axis</b>           | Telomeres in the 25th percentile; TERT rs10069690 T/T; TERT rs2736100 C/C (glioma in the 91st percentile, more prone to clonal expansions); TCL1A G/G.                                                                                                                  | Weakness        | Less replicative reserve of stem cells; the opposite side of TERT increases some cancers. The effect on life expectancy is small, but it affects several mechanisms.                                                                                   | Without telomerase activators; reduce wear (inflammation, oxidative stress, sleep); annual blood count and clonal hematopoiesis test to be discussed (see "what the chip does not see"). <i>suggestion (not applied)</i>                                                                                                                                                                                                                                             |
| <b>12 Bacterial mucosal infections</b>            | Pneumonia in percentile 84.2, urinary infection in 93.8, otitis in 85.3, zone in 74.9; FCGR2A H131R heterozygous (less efficient IgG2 optimization); MBL and TLR4 normal.                                                                                               | Moderate        | Pneumonia is one of the main causes of death in old age; rapamycin and dapagliflozin (urinary infections) are added to the genetic trend.                                                                                                              | Vaccines in day: pneumococcal conjugated (discuss it already, by rapamycin and the score), annual flu, recombinant zone and COVID-19/RSV according to age; hydration and attention to urinary symptoms with dapagliflozin. <i>suggestion (not applied)</i>                                                                                                                                                                                                           |
| <b>13 Glaucoma and tinnitus (eyes and ears)</b>   | Open angle glaucoma in percentile 97.9 (127 variants); CDKN2B-AS1 rs4977756 A/A, SIX6 H141N C/C, TXNRD2; strong genetic myopia. Tinnitus in 89.7, severity in 94.6, loci in 99.8; use of hearing aids in 90 (typical presbycusis).                                      | Moderate-strong | Glaucoma is the main cause of irreversible blindness and evolves without symptoms; untreated hearing loss is one of the major modifiable factors of cognitive decline. They do not shorten life, but they count very much in the late quality of life. | Annual ophthalmologic examination with intraocular pressure, visual field and OCT of the optic nerve; protect the ears from noise (protectors, 60/60 rule on the headphones) and audiogram basis. <i>suggestion (not applied)</i>                                                                                                                                                                                                                                    |
| <b>14 Risk Pharmacogenetics</b>                   | CYP2C9 *1/*3 + VKORC1 T/T (varfarin and NSAIDs), slow NAT2, normal CYP1A2 (*1F A/C), UGT1A1 *28, CYP3A5 *1/*3 (rapamycin); CYP2D6 without copy number.                                                                                                                  | Strong          | Adverse drug effects are a preventable cause of damage; warfarin and long-lived NSAIDs are the most relevant.                                                                                                                                          | Register the genotypes in the clinical process (pharmacological card); NSAIDs at the minimum dose and avoid piroxicam and tenoxicam; if it is necessary to anticoagulate, prefer a direct anticoagulant or dose guided by the genotype. <i>suggestion (not applied)</i>                                                                                                                                                                                              |

|                                                                         |                                                                                                                                                                                   |          |                                                                 |                                                                                                                                                                                               |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>15 Lipids: High CETP, low HDL, triglycerides and polygenic Lp(a)</b> | CETP high activity (3 SNPs, a single signal), HDL genetic in the 17-20 percentiles, triglycerides in 68-73, Lp(a) polygenic in 75 (without strong alleles; kringle IV-2 repeats). | Weakness | In addition to 9p21 atherosclerotic risk, none is strong alone. | Measure Lp(a) once (the chip does not read the size of the gene); ApoB and low triglycerides with fiber, omega-3 at the dose of the plan and little sugar.<br><i>suggestion (not applied)</i> |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## The most favourable 10 genetic points

The 10 most favourable genetic points for longevity. They are the basis on which the plan acts: none dispenses with usual care.

| #  | Point                                                 | Genetics                                                                                                                                                                                                                                                                | Evidence                  |
|----|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| 1  | <b>APOE ε3/ε3, TOMM40 and MAPT H2/H2</b>              | Without ε4 (the largest genetic factor of mortality and Alzheimer's); TOMM40 rs2075650 A/A; MAPT H2/H2 (less Parkinson's and tauopathies).                                                                                                                              | Strong                    |
| 2  | <b>FOXO3 longevity homozygous and low IGF-1 axis</b>  | FOXO3 with 4 alleles of longevity in homozygosity; IGF-1 genetic in the 7th percentile; GHR d3 with 1 copy; IGF1R rs2229765 G/A.                                                                                                                                        | Strong (FOXO3) / moderate |
| 3  | <b>Longevities scores</b>                             | Longevity (330 variants) in the 92nd percentile; score of parental life expectancy in 16 (less alleles that shorten life); HLA-DQA1rs34831921 with 1 longer parental life allele.                                                                                       | Moderate                  |
| 4  | <b>Heart: low polygenic risk</b>                      | Coronary disease (metaGRS) in the 5th percentile, polygenic LDL in 20, atrial fibrillation in 2, stroke in 14-19, dilated cardiomyopathy in 2, resting heart rate in 4; without strong Lp(a) alleles, without factor V Leiden.                                          | Moderate-strong           |
| 5  | <b>Low glucose and glycation</b>                      | HbA1c genetics in the 7-8 percentiles, fasting glucose in 9.5, fasting insulin in 5 (12 variants).                                                                                                                                                                      | Moderate                  |
| 6  | <b>Cancers with very low polygenic risk</b>           | Prostate in percentile 0.9, kidney in 0.8, lymphomas in 1, squamous cell carcinoma in 3, genome melanoma in 6.5 0 variants P/LP plausible in 120 genes; without the founder variants read (BRCA1, CHEK2, APC, HOXB13). Exception: colorectal cancer (see unfavourable). | Moderate                  |
| 7  | <b>Immunity and clean self-immunity</b>               | Clean autoimmune HLA (without B*27, C*06:02, DRB1*15:01, DQ2, DQ8); inflammatory bowel disease in 2-9, celiac disease in 4, psoriasis in 2, asthma in 4 CRP genetics in 12 and IL6R Asp358Ala COVID-19 severe in 4.7, MBL without deficiency and normal TLR4.           | Moderate-strong           |
| 8  | <b>Rim</b>                                            | Genetic renal function in percentile 94, UMOD with 1 protective haplotype, APOL1 without G1, ABCG2 Q141K absent.                                                                                                                                                        | Moderate                  |
| 9  | <b>Blood: low predisposition to clonal expansions</b> | Without JAK2 46/1, SH2B3 R262 C/C, CHIP score in 30, RDW genetic in 3.9 (high RDW predicts mortality).                                                                                                                                                                  | Moderate                  |
| 10 | <b>Tobacco and additions</b>                          | "I have smoked" in the 2nd percentile, a common factor of addictions in 20, a load of alcohol loci in 7.5 and tobacco in 10 (CHRNA5 only has nicotine).                                                                                                                 | Moderate                  |

Also favourable, with less effect: antioxidant regulation (NRF2, NQO1, GCLC, SOD2), mitochondria (PGC-1α Gly482Gly), common DNA repair, Parkinson in the 8th percentile, fluid intelligence in the 79, high genetic vitamin D, osteoarthritis in the 16, vertigo and Ménière disease in the 0.2 loci and SOX17 with fewer risk alleles of arterial pulmonary hypertension.

# What the chip doesn't see

Everything in this report comes from a genotyping chip (about 600 thousand positions) completed by imputation. There are four types of information left out that only the sequencing reads.

## 1. Genome complete to 30x (germinal)

| What is missing                                        | Why does it matter here                                                                                                                                                                                                                   |
|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>GSTM1 and GSTT1 deletions</b>                       | The null GSTM1 is only "probable" and the GSTT1 is not testable; both count on detoxification, which is already the weak point of the II phase (slow NAT2, GSTA1*B/*B).                                                                   |
| <b>Rare variants of hereditary disease</b>             | Only 21% of the P/LP variants known in 81 ACMG genes are read (35% of those with $\geq 2$ stars); the large deletions (BRCA1, MSH2/EPCAM, LDLR) and the new variants are not seen. A negative result does not exclude hereditary disease. |
| <b>HLA</b>                                             | There are only imputed markers (B*58:01, B*57:01 and A*31:01 probable negative); the HLA typing of 2 fields gives certainty before allopurinol, abacavir or carbamazepine and completes the autoimmune profile.                           |
| <b>Complete mitochondrial DNA</b>                      | Only classical 4 variants and haplogroup were read; heteroplasma and rare variants require sequencing.                                                                                                                                    |
| <b>Structural variables and repetitions</b>            | Number of copies of CYP2D6, kringle repetitions IV-2 of LPA (the main determinant of Lp(a)), APOL1 G2, SMN1, alpha-thalassemia, polyT of TOMM40, repetition CAG of the androgen receptor and expansions (C9orf72, HTT, FMR1).             |
| <b>Variants of rare centenarians</b>                   | centSIRT6 (N308K/A313S), ANGPTL8 Q121X and most protective function losses of APOC3, ANGPTL3 and PCSK9 are not read (from the 343 variants of ClinVar in these genes, only 19 were read).                                                 |
| <b>BMPR2 and arterial pulmonary hypertension genes</b> | Of the ~1970 pathogenic variants known in BMPR2 and in more 11 related genes, the chip only covers a small fraction: BMPR2 is not evaluated.                                                                                              |

## 2. Clonal hematopoiesis (CHIP)

CHIP are clones of blood cells with acquired mutations (mainly DNMT3A, TET2, ASXL1, JAK2 V617F). It appears in 10–20% of people over the age of 70, almost doubles coronary disease and increases mortality and the risk of blood cancer. It is somatic: the chip reads the inherited predisposition (typical to low: without JAK2 46/1, SH2B3 C/C, CHIP score in the 30th percentile; TCL1A G/G and TERT pulling upwards), but not the clones. To see it, it is necessary to sequence the blood with a panel of high-depth CHIP mutations (detector clones from ~1–2% of cells). With 9p21 homozygous, a clone TET2 or DNMT3A would add vascular risk.

## 3. Other measurements that the inherited DNA does not replace

- Epigenetic clocks measured in blood (the real biological age, instead of the inherited loci).
- Length of the measured telomeres (the genetic score only gives the trend).
- Lp(a) measured once in life.
- Somatic mutations in other tissues and loss of Y in the blood.

# Mechanisms of ageing

The table below binds only the results of part A (new variables and longevity scores) to the mechanisms. The full map of 84 mechanisms, with all reports, is in the "Map of 84 aging mechanisms" section.

| Mechanism | Unfavourable | Favourable | Typical |
|-----------|--------------|------------|---------|
|-----------|--------------|------------|---------|

## Practical part

Ideas that come out of the synthesis, by order of unfavourable ranking, crossed with the genetic plan v2 and with the current medication (retatrutide, rapamycin, dapagliflozin, telmisartan, tadalafil, lithium 5 mg, naltrexone in low dose, isotretinoin). Nothing was applied; everything is to discuss with the doctor.

### 1. LDL and ApoB very low throughout life

Genetics: 9p21 homozygous (5 SNPs, 4 on chip and 1 imputed); High CETP; HDL p17-20; Lp(a) polygenic p75

With 9p21 in homozygosity, the best modifiable compensation is low exposure time to ApoB. Targets to discuss: LDL < 70 mg/dL and ApoB < 60-70 mg/dL; measure the Lp(a) once; coronary calcium score to calibrate the intensity. Zero tobacco and regular training.

V2 genetic plan and medication: Coherent with the plan (low lipids); concrete targets are for the doctor. • *suggestion (not applied)*

### 2. Liver and lung: protecting a PiMZ liver

Genetics: SERPINA1 PiMZ; PNPLA3 I148M; MBOAT7; HFE H63D; ALT p87; emphysema p96

Alcohol zero or almost zero, low liver fat (retatrutide helps), without smoke or vapors. Base and periodic liver elastography (FibroScan); ALT, GGT and ferritin regular, especially with isotretinoin and rapamycin; ferritin below 100 without supplementary iron.

V2 genetic plan and medication: Compatible with the plan; elastography would be an addition. • *suggestion (not applied)*

### 3. Mood, attention and sleep

Genetics: P95 Depression; ADHD p94-97; insomnia p89; bipolar p79

Regular monitoring of mood and sleep; cognitive-behavioral therapy for insomnia as the first line; morning light, regular time and dark room; melatonin 0.3-1 mg 1-2 h before bedtime if necessary. Low-dose lithium is consistent with this profile.

V2 genetic plan and medication: Lithium 5 mg already in the plane; low dose melatonin to be discussed. • *suggestion (not applied)*

### 4. Thrombosis: prevention in risk situations

Genetics: Venous thrombosis p84-86; fibrinogen p92; F11 T/T

Inform the surgeon about the genetic risk before any surgery or hospitalization (mechanical and pharmacological prophylaxis); in trips of more than 4 h, standing and walking, hydrating and compression stockings. Know the alarm signs.

V2 genetic plan and medication: It is not in the plan; it is a situation rule. • *suggestion (not applied)*

### 5. Aortic aorta and valve

Genetics: P79 ascending aorta (FBN1, ULK4); p95 aortic stenosis; 9p21 (abdominal aneurysm)

Base echocardiogram (aortic and valve diameter) and ultrasound of the abdominal aorta later, to be discussed; avoid fluoroquinolones when there is an alternative.

V2 genetic plan and medication: Accretion to the plan (exams). • *suggestion (not applied)*

### 6. Lean mass and strength

Genetics: Appendicular lean mass p10; force loci p2,4; fragility p75-82; genetic physical activity p7-14

Strength training 3 times a week with progression (the muscle is the organ that most compensates these genes), protein ~140 g/day in 3-4 meals, daily creatine; DEXA base and annual to follow the lean mass during the retatrutide.

V2 genetic plan and medication: Similar to the plan (protein ~140 g/day, creatine); DEXA would be an addition. • *suggestion (not applied)*

### 7. Brain: small pots and reserve

Genetics: White substance p92; PMF1, SLC13A3; Alzheimer without APOE p91; APOE ε3/ε3

Aerobic exercise (150-300 min/week), sleep of 7-8 h, LDL and low glucose, omega-3 at the dose of the plan (3.2-4 g/day of EPA+DHA), cognitive and social stimulus; cognitive assessment based on discussion.

V2 genetic plan and medication: Omega-3 already in the plane. • *suggestion (not applied)*

## 8. Colonoscopy without delay

Genetics: Colorectal cancer p94,3/p78,5; early onset p82,3; polyps (loci) p92,7; without APC/MUTYH/CHEK2

First quality colonoscopy at the age of 45 (or already, if this age has passed), in a center with good rate of detection of adenomas and good preparation, plus FIT (immunochemical test of occult blood in feces) at intervals; the next interval is decided by the findings. The complete scores replace the reading of §24 that genetics did not justify anticipating screening. High fiber, low processed meat, zero alcohol and physical activity reduce the risk.

V2 genetic plan and medication: Accretion to the plan (tracking). • *suggestion (not applied)*

## 9. Vaccines and infections

Genetics: Pneumonia p84,2; urinary infection p93,8; otitis p85,3; zone p74,9; FCGR2A H/R; rapamycin; dapagliflozin

Conjugated pneumococcal vaccine (to discuss it already, by rapamycin and the score), annual flu, recombinant zone and COVID-19/RSV according to age. With dapagliflozin, good hydration and looking for the doctor early in the face of urinary or genital symptoms.

V2 genetic plan and medication: Accretion to the plan (vaccines). • *suggestion (not applied)*

## 10. Aromatic amines and bladder

Genetics: Slow NAT2; Probable null GSTM1; Low PON1; bladder loci p86–95

Avoid carbonized, smoked and highly processed meat, smoke from any origin and permanent hair dyes with aromatic amines; drink water throughout the day; urine analysis (blood in urine) in routine.

V2 genetic plan and medication: Compatible with the plan. • *suggestion (not applied)*

## 11. Eyes and ears

Genetics: Glaucoma p97, 9; SIX6; CDKN2B-AS1; myopia; tinnitus p89,7 (loci p99,8); auditory apparatus p90

Annual ophthalmologic examination with intraocular pressure, visual field and OCT of the optic nerve. Protect the ears from noise (protectors in concerts, works and tools; 60/60 rule on headphones); audiogram base. Communicating tinnitus or sudden hearing loss (a rare warning of tadalafil; isotretinoin rarely causes tinnitus).

V2 genetic plan and medication: Accretion to the plan. • *suggestion (not applied)*

## 12. Complete genome and clonal hematopoiesis

Genetics: ACMG 21% coverage; GSTM1/GSTT1, HLA, Lp(a) KIV-2, CYP2D6 not read; CHIP only in blood

A complete 30× genome of clinical grade, with report of secondary ACMG and pharmacogenetic findings (CYP2D6 with copy number, HLA), closes most of what the chip does not see. A panel of CHIP in the blood (DNMT3A, TET2, ASXL1, JAK2) from the 40–50 years, repeating if positive, gives the somatic state.

V2 genetic plan and medication: Accretion to the plan. • *suggestion (not applied)*

## 13. Pharmacogenetic card

Genetics: CYP2C9 \*1/\*3; VKORC1 T/T; Normal CYP1A2 (\*1F A/C); UGT1A1 \*28; CYP3A5 \*1/\*3; Slow NAT2

To put the genotypes in the clinical process: low-dose and guided warfarin (or direct anticoagulant), NSAIDs at the minimum dose and without piroxicam or tenoxicam, care with isoniazid and sulfonamides (NAT2), caffeine with moderation (CYP1A2).

V2 genetic plan and medication: Accretion to the plan. • *suggestion (not applied)*

## Avoiding or with caution

- Telomerase activators (TA-65, cycloastragenol): his TERT axis already has the side of cancer (p91 glioma). *suggestion (not applied)*
- Growth hormone secretagogues, IGF-1 and leucine or BCAA isolated: contradict the favourable profile of FOXO3 and low IGF-1 and rapamycin. *suggestion (not applied)*
- Nicotine in any form (tobacco, vaping, bags): 9p21, emphysema p96, PiMZ and CHRNA5. *suggestion (not applied)*
- Regular alcohol: PiMZ + PNPLA3, high liver enzymes loci and beverages/week p95. *suggestion (not applied)*
- Long half-life NSAIDs (pyroxicam, tenoxicam, meloxicam in high doses) with CYP2C9 \*3; fluoroquinolones when there is an alternative (aorta). *suggestion (not applied)*
- Iron without proven deficit (HFE H63D and PiMZ liver). *suggestion (not applied)*
- Strict ketogenic diet with dapagliflozin (ketoacidosis). *suggestion (not applied)*

## Limitations

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- The synthesis combines reports made with the same chip and the same imputation: the reading errors of a genotype propagate to all reports that use it.
- The state of each mechanism is a qualitative judgment on the known genes; many mechanisms have few described genes and none has a validated score. The ranking is a clinical ordering, not a calculation of years of life.
- The polygenic scores explain a small fraction of the variation (the longevity, less than 5%) and the percentiles come from a reference of 633 genomes; a high percentile is an inclination, not an individual prediction.
- FOXO3 and GHR d3 replicate poorly outside the original cohorts; the effect of GHR d3 on life expectancy was seen mainly in d3/d3.
- Protective function losses are rare: absence is the expected result and is not unfavourable; many are not in the chip (only 19 of 343 read).
- The chip does not read structural variants, repetitions, integer deletions (GSTM1/GSTT1), complete HLA, mitochondrial heteroplasma or somatic mutations (CHIP).
- The integrated reports after this synthesis are not reflected in the map.

## Methods and sources

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**Genotypes:** first the genotyping chip (GRCh38, strand +), by rsID and, for cast rsIDs, by the position; then the local imputation (Beagle 5.5 with a public reference panel of complete genomes), only with  $DR2 \geq 0.8$ . The variants that do not pass are "SNP not found in the data". The raw DNA was processed locally; to the public databases only rsIDs were sent.

**Variation classification:** The number of copies of the unfavourable allele is counted and compared to what was expected in the reference population: at least 0.5 copies above that expected = unfavourable; at least 0.5 below = favourable; the rest = typical/intermediate. For rare variants of large effect, having 0 copies = favourable and having  $\geq 1$  = unfavourable. The unfavourable alleles come from the literature (cured) or, when indicated, from the strongest association in GWAS Catalog.

**Main variants per gene:** for genes labeled as "GWAS Catalog", the variant with the strongest genomic association ( $p \leq 5 \times 10^{-8}$ ) is used, mapped to this gene and with available genotype.

**Polygenic scores (PGS Catalog):** Harmonized coefficients for GRCh38 applied to the imputed genome (chip +  $DR2 \geq 0.8$ , dosage). Empirical percentile in the reference population ( $n=633$ ), calculated with the same variants. List: PGS000906 (PGS Catalog; Clumping and Thresholding (C+T); Longevity; PGP000237, PMID 33216869 · PGS002795 (PGS Catalog; Genome-wide signifner SNPs; Longevity (>90th survival percentile); PGP000378, PMID 34091660 · PGS001393 (PGS Catalog; snpnet; Father's acts at death; PGP000244, PMID 35324888 · PGS000319 (PGS Catalog; Composite PRS (component scores combined by  $\log(HR)$ ); All-cause mortality (male); PGP000095, PMID 32758451).

**GWAS Catalog loci scores:** for traits without PGS, weighted sum of risk variants ( $p \leq 5 \times 10^{-8}$ , one per window of 500kb), with a percentile calculated in the same way.

**Percentis:**  $\leq 25$  or  $\geq 75$  (in the direction of the trait) = favourable or unfavourable; between 25 and 75 = typical.

### Main sources

- Willcox BJ et al. FOXO3A genotype is strongly associated with human longevity (PNAS 2008).
- Timers PRHJ et al. Genomics of 1 million parent lifespans implicates novel paths and common diseases and distinguishes survival chances (eLife 2019).
- Pilling LC et al. Human longevity: 25 genetic loci associated with parental lifespan (Aging 2017).
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